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1. INTRODUCTION

This report outlines the functional and non-functional requirements for a mobile application designed to manage student attendance using biometric authentication. The app aims to improve efficiency, accuracy, and transparency in attendance tracking within a school environment.

2. REQUIREMENT ANALYSIS

Requirement analysis is a crucial initial phase in software development and systems engineering. It's the process of gathering, analyzing, documenting, and validating the specific needs and expectations of a project or product.

3. Functional Requirements

They define the specific actions or functionalities a system or application must perform. They outline what the system should do under various conditions and how it should process inputs to deliver the expected outputs.

3.1 Core Functionalities

1.User Management

- The app shall allow users to register for an account using a valid school ID and email address.
- The app shall implement multi-factor authentication for login, requiring both username/password and biometric verification (fingerprint, facial recognition).
- The app shall define user roles (student, teacher, parent/guardian) with appropriate access controls.
- The app may allow users to manage basic profile information (name, photo, contact details) within their role.

2. Attendance Management

- The app shall enable students to securely mark their attendance using a registered biometric identifier (fingerprint, facial recognition).
- The app shall allow configuration of a specific attendance window (timeframe) for each class or the entire school.
- (Optional): The app may capture geolocation data during attendance marking with user consent.
- The app shall allow students to mark attendance offline, with data syncing automatically upon regaining internet connection.
- The app shall provide students with access to their own attendance history for viewing purposes.

3. Data Management

- The app shall securely store all attendance data on the cloud, adhering to data privacy regulations (GDPR, CCPA).
- The app shall restrict access to attendance records based on user roles (students can only see their own, teachers can see their class data, etc.).
- The app shall offer advanced filtering and search functionalities for attendance data (date, class, student name).
- The app shall enable users to export attendance reports in various formats (PDF, CSV, Excel) for further analysis.

4. Additional Features

- The app may provide teachers with a real-time view of student attendance data for their classes, including student details (names, photos).
- The app may allow authorized teachers to mark attendance for absent students with reason selection and notification sent to parents.

- The app may offer teachers access to historical attendance data for their classes for trend analysis.
- The app may provide parents/guardians with a view of their child's attendance history, including details like date, time, and location (if enabled).
- The app may enable parents/guardians to receive push notifications for absences, tardiness, or early leave of their child.
- The app may allow parents/guardians to submit leave requests for their child with reason selection and an approval workflow.

5. Reporting

- The app shall generate comprehensive attendance reports for the school administration with various options (daily, weekly, monthly, yearly).
- The app shall allow filtering of attendance reports based on class, grade level, and date range for specific analysis.
- The app shall facilitate tracking of student attendance patterns to identify potential issues like absenteeism and tardiness.
- The app may generate reports on leave requests submitted by parents/guardians.

6. Integration

- The app may integrate with existing school management systems for data synchronization (if applicable).

These functional requirements provide a detailed roadmap for the core functionalities and optional features of the biometric school attendance mobile application.

4. Non-Functional Requirements

4.1 Performance:

- Responsiveness: The app should be responsive and function smoothly on various mobile devices (smartphones, tablets) with different operating systems (Android, iOS). Load times for screens and data retrieval should be minimal.

- Offline Functionality: Offline attendance marking should be efficient with minimal impact on device battery life. The app should be usable even without an internet connection in areas with weak or no signal. Data should be stored locally and synced automatically upon internet availability.
- Data Syncing: Data syncing between the device and the cloud server should occur quickly upon internet connection availability. Efficient data transfer protocols should be used to minimize bandwidth usage.

4.2 Usability:

- User Interface (UI): The user interface should be intuitive, user-friendly, and visually appealing for all user groups (students, teachers, parents/guardians). The layout should be clear, with easy navigation and minimal learning curve.

- Localization: The app should be available in multiple languages to cater to diverse school populations. Language options should be easily accessible within the app settings.
- Accessibility: Accessibility features should be incorporated to support students with visual impairments (screen reader compatibility, high contrast mode), learning disabilities (larger fonts, text-to-speech options), and other accessibility needs.
- Help and Tutorials: Contextual help and tutorials should be available within the app for user guidance. This can include tooltips, FAQs, or in-app guides explaining functionalities.

4.3 Reliability:

- Uptime: The app should function reliably with minimal downtime or errors. The system should be designed with high availability in mind to ensure continuous operation.
- Data Integrity: Data synchronization should be consistent and accurate to ensure data integrity. Error handling mechanisms should be implemented to catch and correct any discrepancies during data transfer.
- Updates and Bug Fixes: Regular updates and bug fixes should be released to maintain optimal performance, address user feedback, and patch any security vulnerabilities. A clear update notification system should be in place to inform users about new versions.

4.4 Scalability:

- User Base: The app should be designed to scale and accommodate a growing user base. The system architecture should be able to handle an increasing number of students, teachers, and parents using the app concurrently.
- Data Storage: The cloud storage solution should be scalable to accommodate the growing volume of attendance data generated over time.

4.5 Security:

- Data Encryption: Secure data encryption for attendance records at rest and in transit is critical (using industry standards like AES-256). This ensures that even if data is intercepted, it remains unreadable.
- User Authentication: Multi-factor authentication for login (username/password + biometric) should be implemented to prevent unauthorized access. Biometric data should be securely stored using hashing algorithms and not stored as raw data.
- Authorization: User access control based on roles with the least privilege principle should be implemented. This means users only have access to the data and functionalities relevant to their role (student, teacher, parent/guardian).
- Security Audits: Regular security audits and penetration testing should be conducted to identify and address potential vulnerabilities in the app and server infrastructure.

- Data Privacy: The app should comply with relevant data privacy regulations (e.g., GDPR, CCPA) regarding student data collection, storage, and usage. Transparent privacy policies should be available within the app.

4.6 Interoperability:

- Existing Systems: The app should be designed to integrate with existing school management systems (if applicable) for seamless data exchange (e.g., student information, class schedules). APIs or data import/export functionalities can be implemented for this purpose.

4.7 Maintainability:

- Code Structure: The app code should be well-documented, modular, and easy to understand for future maintenance and development.
- Error Reporting: Robust error reporting mechanisms should be implemented to identify and troubleshoot any issues within the app.
- Configurability: The app should be configurable to adapt to different school needs. This could include customizable attendance windows, notification settings, and user roles.

By considering these non-functional requirements, the development team can create a reliable, scalable, and user-friendly mobile application that effectively manages student attendance using biometrics.

5. Other Requirements

5.1 Hardware Requirements:

- Device: Any Android compatible Device.
- Finger-print device: MFS100.

5.2 Software Requirements:

- 2.2.1 Front End: JavaScript
- 2.2.2 Database: MongoDB / MYSQL

5.2.3. Software Tool:

- Android studio

CONCLUSION

By considering these requirements, the development team can create a reliable, scalable, and user-friendly mobile application that effectively manages student attendance using biometrics.